Software Requirements Specification (SRS)

Project: Task Manager Application

Team: Team 6

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Technology Stack: JavaScript (React.js + Node.js) / Python (Django)

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Revision history

Version	Date	Author	Change summary	Approval
1.0	03-09-2026	Team 6	Initial SRS drafted for Task Manager Application	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

Table of Contents

1. Introduction
2. Overall description
3. External interfaces
4. System features (detailed)
5. Non-functional requirements (detailed)
6. Quality attributes & Acceptance tests
7. UML Use-Case Diagram
8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for the Task Manager Application developed by Team 6. It defines the functional and non-functional requirements, interfaces, and verification criteria for a web-based application that lets individuals and teams create, organize, and track work items to completion.

1.2 Scope

Covers user registration and authentication, task creation and lifecycle management, categorization and prioritization, notifications and reminders, team collaboration (sharing and assignment), and progress reporting. The application will be built with a React.js front end, and a backend implemented in either Node.js or Python (Django), exposing a REST API. Excludes payment/billing systems and third-party calendar synchronisation, which are out of scope for version 1.0.

1.3 Audience

Developers, QA Engineers, Course Instructors/Evaluators, and prospective end users (individuals and small teams managing daily tasks).

1.4 Definitions

List of acronyms: UI (User Interface), API (Application Programming Interface), REST (Representational State Transfer), JWT (JSON Web Token), CRUD (Create, Read, Update, Delete), DB (Database), RBAC (Role-Based Access Control), TLS (Transport Layer Security), UX (User Experience), SPA (Single Page Application).

2. Overall description

2.1 Product perspective

The Task Manager Application is a standalone, self-contained web application consisting of a React.js single-page front end, a REST API backend (Node.js/Express or Python/Django), and a relational database. It is not dependent on any existing legacy system and integrates only with an email service (for notifications and password resets) and, optionally, an OAuth provider for social login.

2.2 Major product functions (detailed)

- Register and authenticate users
- Create, edit, and delete tasks
- Set task priority, category/tags, and due dates
- Track task progress through status states (To-Do, In Progress, Completed)
- Organize tasks into projects/boards with filter, sort, and search
- Send in-app and email notifications/reminders
- Invite team members, assign tasks, and comment collaboratively
- Provide dashboards and productivity reports

2.3 User roles and characteristics (expanded)

- User: registers an account, creates and manages their own tasks, and collaborates on shared projects; expects a fast, intuitive interface.
- Team Member: a user who is invited to a shared project and may be assigned tasks, comment, and update task status.
- Admin: manages user accounts, monitors platform-wide usage, and reviews aggregate analytics and audit logs.

2.4 Operating environment

Modern web browsers (Chrome, Firefox, Edge, Safari) on desktop and mobile; backend hosted on a cloud application server; managed relational database (e.g., PostgreSQL/MySQL); deployed behind HTTPS with a reverse proxy/load balancer.

2.5 Constraints

Must use React.js for the front end and either Node.js or Python (Django) for the backend, per the project brief. Must be delivered as a REST API. Password storage and data-in-transit must follow standard security practice (see Section 5.1). Development timeline is constrained to a single academic semester.

3. External interface requirements

3.1 User interfaces

Primary UI: a responsive single-page web application with a dashboard, task list/board views, and modal forms for task creation and editing. Accessibility: keyboard-navigable forms and sufficient colour contrast for status/priority indicators.

3.2 Hardware interfaces

None beyond standard client devices (desktop, laptop, tablet, or smartphone) capable of running a modern web browser.

3.3 Software interfaces

- Backend REST API (JSON over HTTPS) for all task, user, and project operations.
- Email service API (e.g., SMTP/SendGrid) for notifications, reminders, and password-reset links.
- Optional OAuth provider (e.g., Google) for social login.
- WebSocket channel for real-time notifications and comment updates.

3.4 Communications

- TLS 1.2+ enforced for all client-server and server-database traffic.
- REST API responses returned as JSON with standard HTTP status codes; retry with exponential backoff on transient network failures.

This SRS defines 24 functional requirements (Section 4), 5 non-functional requirements (Section 5), 2 security objectives, and 5 security requirements (Section 5.1), satisfying the minimum coverage required for this project.

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow TM-F-###.

4.1 Authentication & Account Management

Description: Register, authenticate, and secure user accounts.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-001	The system shall allow a new user to register using name, email, and password.	Functional	High	Product Owner	AC-TM-F-001: Valid registration creates an account and sends confirmation. Test: TC-Auth-01	Email must be unique
TM-F-002	The system shall authenticate registered users via email and password before granting access.	Functional	High	Security	AC-TM-F-002: Correct credentials log the user in and issue a session token. Test: TC-Auth-02	JWT-based session
TM-F-003	The system shall hash and salt user passwords before storage; plaintext passwords shall never be stored.	Functional	High	Security	AC-TM-F-003: Database inspection shows only hashed values. Test: TC-Auth-03	bcrypt/argon2
TM-F-004	The system shall allow a user to reset a forgotten password via a time-limited email link.	Functional	Medium	User	AC-TM-F-004: Reset link expires after 30 minutes and updates password once used. Test: TC-Auth-04	Email service dependency
TM-F-005	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Functional	High	Security	AC-TM-F-005: 6th attempt is rejected with lockout message. Test: TC-Auth-05	Audit log entry required

4.2 Task Creation & Lifecycle Management

Description: Create, edit, delete, and track the status/priority/category of tasks.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-010	The system shall allow a logged-in user to create a task with a title, description, and due date.	Functional	High	User	AC-TM-F-010: New task appears in the user's task list immediately. Test: TC-Task-01	Title mandatory
TM-F-011	The system shall allow a user to edit the title, description, due date, and status of an existing task.	Functional	High	User	AC-TM-F-011: Edited fields persist after page refresh. Test: TC-Task-02	Only owner/assignee may edit
TM-F-012	The system shall allow a user to delete a task they own, with a confirmation prompt.	Functional	High	User	AC-TM-F-012: Deleted task is removed from list and database. Test: TC-Task-03	Soft delete for 30 days
TM-F-013	The system shall allow a user to mark a task as To-Do, In Progress, or Completed.	Functional	High	User	AC-TM-F-013: Status change updates the dashboard counts. Test: TC-Task-04	-
TM-F-014	The system shall allow a user to set a priority level (Low, Medium, High) for each task.	Functional	Medium	User	AC-TM-F-014: Priority is displayed with a colour indicator. Test: TC-Task-05	Include relationship with TM-F-010
TM-F-015	The system shall allow a user to assign one or more tags/categories to a task.	Functional	Medium	User	AC-TM-F-015: Tasks can be filtered by tag. Test: TC-Task-06	-

4.3 Task Organization

Description: Filter, sort, search, and group tasks into projects/boards.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-020	The system shall allow a user to filter tasks by status, priority, category, and due date.	Functional	Medium	User	AC-TM-F-020: Filtered results match the selected criteria. Test: TC-Org-01	-
TM-F-021	The system shall allow a user to sort the task list by due date, priority, or creation date.	Functional	Medium	User	AC-TM-F-021: List order reflects the chosen sort key. Test: TC-Org-02	-
TM-F-022	The system shall provide a search function that returns tasks matching a keyword in title or description.	Functional	Medium	User	AC-TM-F-022: Search results update within 1 second of query submission. Test: TC-Org-03	-
TM-F-023	The system shall organize tasks into user-defined projects or boards.	Functional	Low	User	AC-TM-F-023: Tasks can be moved between boards via drag-and-drop. Test: TC-Org-04	Kanban-style board view

4.4 Notifications & Reminders

Description: Keep users informed of upcoming deadlines and assignments.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-030	The system shall send an in-app notification when a task is due within 24 hours.	Functional	Medium	User	AC-TM-F-030: Notification appears in the notification centre. Test: TC-Notif-01	Background scheduler
TM-F-031	The system shall send an email reminder for tasks marked as High priority that are overdue.	Functional	Medium	Operations	AC-TM-F-031: Overdue high-priority tasks trigger an email within 10 minutes. Test: TC-Notif-02	Email service dependency
TM-F-032	The system shall notify a user in real time when they are assigned a new task.	Functional	Medium	User	AC-TM-F-032: Assignee receives a push/WebSocket notification within 5 seconds. Test: TC-Notif-03	WebSocket channel required

4.5 Collaboration & Sharing

Description: Invite team members, assign tasks, and comment on shared work.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-040	The system shall allow a task owner to invite another registered user to a project by email.	Functional	High	User	AC-TM-F-040: Invited user gains access after accepting the invite. Test: TC-Collab-01	-
TM-F-041	The system shall allow a user to assign a task to another team member.	Functional	High	User	AC-TM-F-041: Assigned task appears in the assignee's task list. Test: TC-Collab-02	Requires TM-F-040
TM-F-042	The system shall allow team members to add comments to a shared task.	Functional	Medium	User	AC-TM-F-042: New comments appear for all project members in real time. Test: TC-Collab-03	-

4.6 Reporting & Progress Tracking

Description: Summarise progress via dashboards and reports.

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
TM-F-050	The system shall display a dashboard summarising task counts by status for the logged-	Functional	Medium	User	AC-TM-F-050: Dashboard counts match the underlying task data. Test:	-

Req ID	Requirement (shall...)	Type	Priority	Source/ Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	in user.				TC-Rep-01	
TM-F-051	The system shall generate a weekly productivity report showing tasks completed versus tasks created.	Functional	Low	Product Owner	AC-TM-F-051: Report totals reconcile with task history. Test: TC-Rep-02	-
TM-F-052	The system shall allow an admin to view aggregate usage analytics across all teams.	Functional	Low	Admin	AC-TM-F-052: Analytics view accessible only to admin role. Test: TC-Rep-03	Role-based access control

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs TM-NF-###.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
TM-NF-001	Page load and API response time shall be ≤ 2 seconds for 90% of requests under normal load (up to 500 concurrent users).	Performance	High	90th percentile ≤ 2s in load test. Test: TC-Perf-01
TM-NF-002	The system shall provide 99.5% monthly uptime, excluding scheduled maintenance windows.	Reliability	High	Uptime monitoring reports ≥99.5% per month. Test: Ops reports
TM-NF-003	All data in transit shall be encrypted using TLS 1.2 or higher; all API endpoints requiring authentication shall reject unauthenticated requests.	Security	High	TLS scan and endpoint audit pass. Test: TC-Sec-01
TM-NF-004	The user interface shall be responsive and usable on screen widths from 360px (mobile) to 1920px (desktop).	Usability	Medium	Manual responsive-design checklist pass across 3 device classes. Test: TC-UX-01
TM-NF-005	The backend shall be horizontally scalable, supporting an increase to 5,000 concurrent users without code changes.	Scalability	Medium	Load test at 5,000 simulated users meets NF-001 targets. Test: TC-Scale-01

5.1 Security

5.1.1 Security Objectives

- Confidentiality: Protect user credentials and task/project data from unauthorized disclosure.
- Integrity: Ensure task and project data cannot be modified by unauthorized users or corrupted in transit.
- Availability & Accountability: Maintain controlled access so only authenticated, authorized users can act on the system, with actions traceable via audit logs.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
TM-SR-001	TLS 1.2+ shall be mandatory for all client-server and server-database network connections.	Security	High	AC-TM-SR-001: TLS scan confirms no plaintext channel. Test: TC-Sec-01
TM-SR-002	User passwords shall be stored using a salted, one-way hash (bcrypt or equivalent); plaintext shall never be persisted or logged.	Security	High	AC-TM-SR-002: Database and log review shows no plaintext passwords. Test: TC-Sec-02
TM-SR-003	All API endpoints shall validate and sanitise user input to prevent SQL injection and cross-site scripting (XSS).	Security	High	AC-TM-SR-003: OWASP ZAP scan reports no critical injection/XSS findings. Test: TC-Sec-03
TM-SR-004	The system shall enforce role-based access control (RBAC) so that a user can only access tasks and projects they own or are a member of.	Security	High	AC-TM-SR-004: Access attempt to another user's private task is rejected with 403. Test: TC-Sec-04
TM-SR-005	The system shall rate-limit authentication endpoints to no more than 10 requests per minute per IP address to mitigate brute-force attacks.	Security	Medium	AC-TM-SR-005: 11th request within a minute is rejected with 429. Test: TC-Sec-05

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and RTM shows all test cases passed.

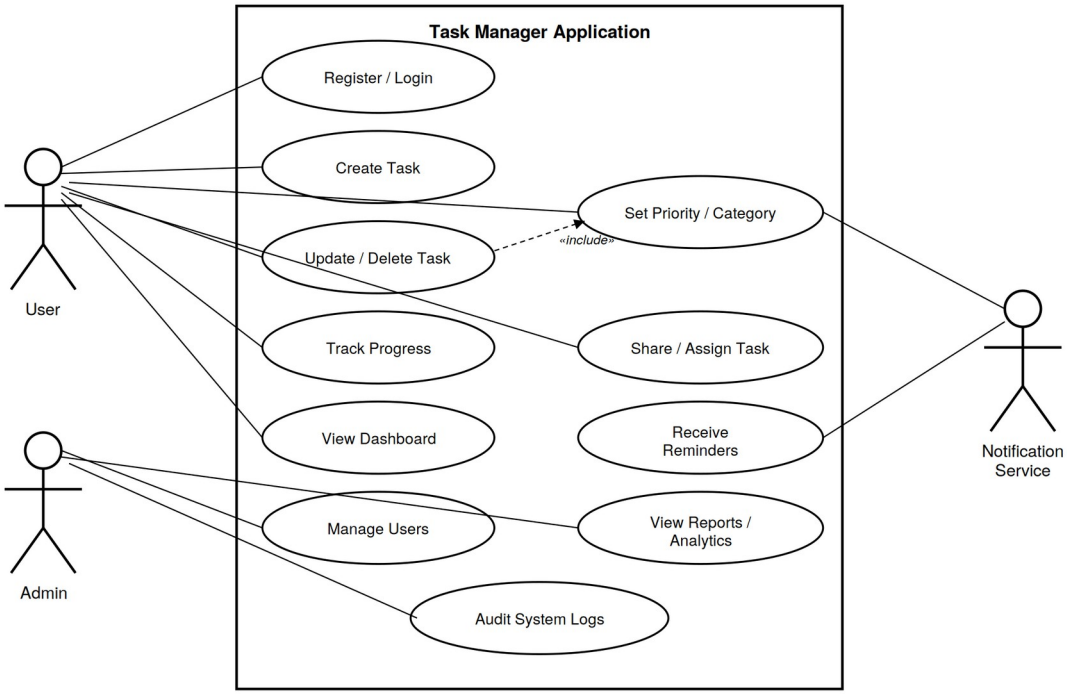
Acceptance test suites: Registration & Authentication, Task CRUD, Organization (Filter/Sort/Search), Notifications, Collaboration, Reporting, Performance, and Security.

7. System models and diagrams

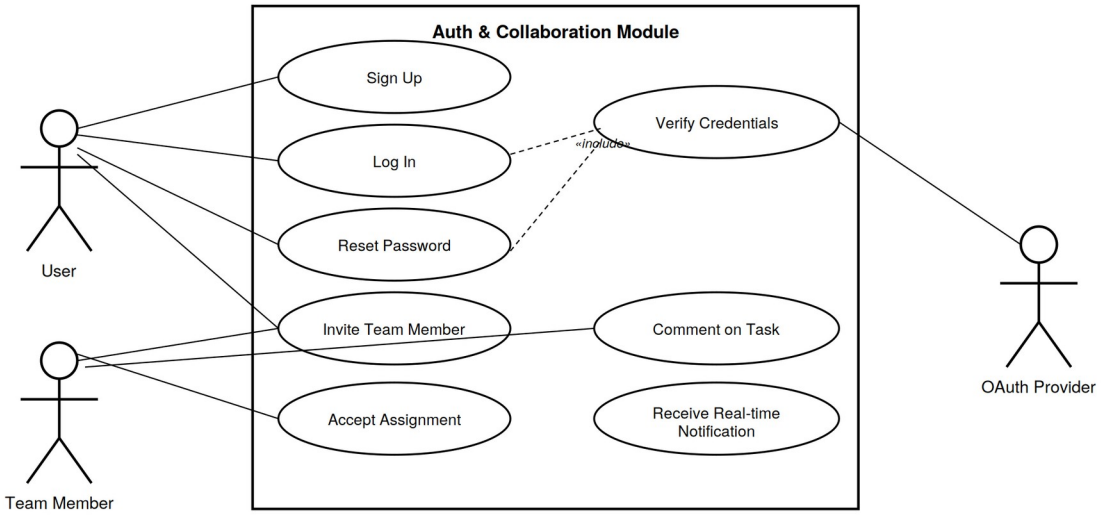
7.1 UML Use-Case diagram

Two use-case diagrams are provided below, covering (1) core task-management functionality and (2) the authentication and collaboration subsystem.

Use Case Diagram 1: Core Task Management (System Boundary)



Use Case Diagram 2: Authentication & Collaboration Subsystem



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
TM-F-001	User registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	-
TM-F-002	User login	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	-
TM-F-010	Create task	4.2 / DS-Task-01	TaskModule	TC-Task-01	N	-
TM-F-011	Edit task	4.2 / DS-Task-02	TaskModule	TC-Task-02	N	-
TM-F-013	Task status update	4.2 / DS-Task-03	TaskModule	TC-Task-04	N	-
TM-F-030	Due-date notification	4.4 / DS-Notif-01	NotificationModule	TC-Notif-01	N	Depends on scheduler
TM-F-041	Task assignment	4.5 / DS-Collab-01	CollabModule	TC-Collab-02	N	-
TM-F-050	Dashboard summary	4.6 / DS-Report-01	ReportModule	TC-Rep-01	N	-
TM-NF-001	Response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	-
TM-NF-003	TLS encryption	5 / DS-Sec-01	API Gateway	TC-Sec-01	N	-
TM-SR-002	Password hashing	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-02	N	-
TM-SR-004	Role-based access control	5.1.2 / DS-Sec-04	AuthModule	TC-Sec-04	N	-